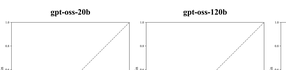

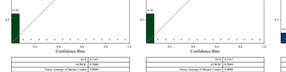

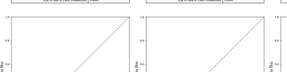

Calibration of Non-Normalized Label Probabilities for Prompt 4 (All Labels)

MMLU

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2167	0.1144	0.2014	0.0600	0.1063	0.1634	0.1043	0.2166	0.2166	0.2167	0.2166	0.2167	0.1986	0.2167	0.2167	0.2166	0.2161	0.2166
AUROC	0.5000	0.5044	0.8339	0.7225	0.8363	0.8038	0.9034	0.8951	0.5588	0.8157	0.8899	0.8082	0.9111	0.9028	0.8701	0.5884	0.8244	0.8577	0.8934
Norm. Entropy of Bucket Counts	0.0000	0.0000	0.5893	0.2021	0.6497	0.5912	0.3728	0.5640	0.0015	0.0018	0.0001	0.0008	0.0002	0.2302	0.0000	0.0000	0.0133	0.0131	0.0008
Invalid Answer Count	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042	0/14042
Accuracy	0.0000	0.0058	0.4814	0.4897	0.4844	0.6016	0.5939	0.5835	0.1050	0.5001	0.6149	0.4665	0.6171	0.5964	0.5378	0.1201	0.5143	0.5922	0.5847
Median of Label Probabilities	0.0000	0.0000	0.0533	0.0158	0.0324	0.0434	0.0030	0.0287	0.0000	0.0000	0.0000	0.0000	0.0000	0.0062	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Label Probabilities	0.0000	0.0000	0.1634	0.0319	0.3158	0.1846	0.0503	0.1772	0.0000	0.0065	0.0000	0.0000	0.0000	0.0309	0.0000	0.0000	0.0000	0.0000	0.0000
Median of Sum of Label Probabilities	0.0000	0.0000	0.5141	0.1221	0.8323	0.5655	0.2448	0.5483	0.0000	0.0156	0.0000	0.0017	0.0000	0.1112	0.0000	0.0000	0.0000	0.0002	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1643	0.0919	0.1437	0.1672	0.2384	0.2658	0.0000	0.0128	0.0000	0.0036	0.0000	0.1037	0.0000	0.0000	0.0001	0.0008	0.0000

ArcReasoning

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2167	0.1326	0.1943	0.0573	0.1241	0.1588	0.1175	0.2167	0.2167	0.2160	0.2167	0.2166	0.1890	0.2167	0.2167	0.2166	0.2163	0.2167
AUROC	0.5000	0.5004	0.9899	0.8664	0.9749	0.9049	0.9951	0.9947	0.6238	0.9699	0.9836	0.9182	0.9855	0.9953	0.9875	0.5037	0.8602	0.9123	0.9908
Norm. Entropy of Bucket Counts	0.0000	0.0000	0.5161	0.2429	0.5307	0.3753	0.4292	0.4948	0.0008	0.0000	0.0068	0.0000	0.0006	0.2946	0.0000	0.0000	0.0123	0.0125	0.0000
Invalid Answer Count	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358	0/3358
Accuracy	0.0000	0.0009	0.7052	0.6828	0.6775	0.7421	0.7439	0.7382	0.2222	0.6894	0.7406	0.6263	0.7362	0.7466	0.7370	0.0048	0.5277	0.6307	0.7362
Median of Label Probabilities	0.0000	0.0000	0.0216	0.0067	0.0046	0.0148	0.0013	0.0077	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Label Probabilities	0.0000	0.0000	0.1912	0.0153	0.3317	0.0461	0.1026	0.2511	0.0000	0.0053	0.0000	0.0000	0.0000	0.0327	0.0000	0.0000	0.0000	0.0000	0.0000
Median of Sum of Label Probabilities	0.0000	0.0000	0.5648	0.1214	0.8138	0.5092	0.4134	0.6448	0.0000	0.0127	0.0000	0.0012	0.0000	0.1425	0.0000	0.0000	0.0000	0.0002	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1223	0.1340	0.1550	0.1944	0.2383	0.2130	0.0001	0.0072	0.0000	0.0015	0.0000	0.0847	0.0000	0.0000	0.0001	0.0009	0.0000

GSM8KMC

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2167	0.1717	0.2152	0.0889	0.0999	0.1958	0.1716	0.2167	0.2167	0.2035	0.2167	0.2164	0.2029	0.2167	0.2167	0.2177	0.2165	0.2167
AUROC	0.5002	0.5147	0.6518	0.5838	0.6907	0.7097	0.7989	0.7438	0.5507	0.5487	0.8000	0.6719	0.8103	0.8918	0.5968	0.5116	0.5990	0.8284	0.8308
Norm. Entropy of Bucket Counts	0.0000	0.0000	0.4153	0.0393	0.8090	0.6901	0.2581	0.4143	0.0001	0.0000	0.1007	0.0000	0.0080	0.1982	0.0000	0.0000	0.0292	0.0312	0.0000
Invalid Answer Count	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468	0/7468
Accuracy	0.0000	0.0020	0.2866	0.3535	0.2895	0.4796	0.3907	0.3851	0.1031	0.0770	0.4464	0.2425	0.4543	0.5216	0.1638	0.0165	0.1727	0.5740	0.4120
Median of Label Probabilities	0.0000	0.0000	0.0531	0.0195	0.1531	0.1233	0.0181	0.0429	0.0000	0.0000	0.0000	0.0000	0.0000	0.0113	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Label Probabilities	0.0000	0.0000	0.0811	0.0245	0.3302	0.2238	0.0464	0.0804	0.0000	0.0000	0.0000	0.0000	0.0000	0.0385	0.0000	0.0000	0.0000	0.0000	0.0000
Median of Sum of Label Probabilities	0.0000	0.0000	0.2930	0.0790	0.9314	0.6450	0.1432	0.2768	0.0000	0.0000	0.0006	0.0005	0.0000	0.1045	0.0000	0.0000	0.0000	0.0001	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.0855	0.0356	0.0435	0.1498	0.1258	0.1642	0.0000	0.0000	0.0140	0.0011	0.0001	0.0863	0.0000	0.0000	0.0000	0.0002	0.0001

GPQA

	gpt-oss-20b	gpt-oss-120b	Ministral-8B-Instruct	Mistral-Nemo-Base	Mistral-Nemo-Instruct	Mistral-Small-3.1-24B-Base	Mistral-Small-3.2-24B-Instruct	Magistral-Small	Magistral-Small-Reasoning	Llama-3.1-70B	Llama-3.3-70B-Instruct	Llama-4-Scout-17B-16E	Llama-4-Scout-17B-16E-Instruct	Qwen3-30B-A3B-Base	Qwen3-30B-A3B-Instruct	Qwen3-30B-A3B-Thinking	DeepSeek-R1-Distill-Llama-70B	DeepSeek-R1-Distill-Qwen-32B	gemma-3-27b-it
ECE	0.2167	0.2167	0.1635	0.1889	0.1205	0.1231	0.1944	0.1709	0.2167	0.2167	0.2169	0.2167	0.2130	0.2037	0.2167	0.2167	0.2167	0.2167	0.2173
AUROC	0.5004	0.5082	0.5659	0.5311	0.5906	0.5706	0.6334	0.6052	0.5007	0.5390	0.6656	0.5594	0.6682	0.6293	0.5723	0.5238	0.5531	0.6241	0.6186
Norm. Entropy of Bucket Counts	0.0000	0.0000	0.4495	0.3132	0.7311	0.6058	0.2481	0.4121	0.0000	0.0000	0.0129	0.0000	0.0527	0.1889	0.0000	0.0000	0.0017	0.0000	0.0050
Invalid Answer Count	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448	0/448
Accuracy	0.0000	0.0022	0.2344	0.3080	0.2879	0.3259	0.3304	0.3214	0.0022	0.0915	0.3571	0.1004	0.3393	0.2589	0.1540	0.0379	0.0781	0.2232	0.2277
Median of Label Probabilities	0.0000	0.0000	0.0712	0.0434	0.1161	0.1104	0.0101	0.0377	0.0000	0.0000	0.0000	0.0000	0.0000	0.0185	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Label Probabilities	0.0000	0.0000	0.0701	0.0658	0.2317	0.1654	0.0305	0.0776	0.0000	0.0000	0.0000	0.0000	0.0000	0.0317	0.0000	0.0000	0.0000	0.0000	0.0000
Median of Sum of Label Probabilities	0.0000	0.0000	0.3485	0.1913	0.7377	0.4572	0.1237	0.2796	0.0000	0.0000	0.0000	0.0000	0.0001	0.1067	0.0000	0.0000	0.0000	0.0000	0.0000
IQR of Sum of Label Probabilities	0.0000	0.0000	0.1319	0.1299	0.1681	0.1572	0.1771	0.2362	0.0000	0.0049	0.0000	0.0004	0.0025	0.1145	0.0000	0.0000	0.0000	0.0000	0.0001